

Cross-Testing Report

Summary of Model Performance

Model_A (Balanced-trained) on Imbalanced Test Set

- Accuracy: 0.5009
- Precision: 0.5115
- Recall: 0.5009
- F1-Score: 0.4939
- Accuracy (0.5009) dropped compared to performance on balanced data, suggesting difficulty handling skewed distributions.
- The macro average F1-score (0.50) shows relatively fair performance across all classes but still reflects class imbalance sensitivity.
- Stronger recall for classes with higher frequency in the imbalanced dataset (Class 5: Recall 0.77).

Confusion Matrix:

```
[[2986 1628 257 46 83]
 [2074 3818 1158 266 184]
 [1125 3695 4261 2443 976]
 [ 350 1095 2977 6316 4262]
 [ 174 188 400 1573 7665]]
```

Model_B (Imbalanced-trained) on Balanced Test Set

- Accuracy: 0.5227
- Precision: 0.5093
- Recall: 0.5227
- F1-Score: 0.5125
- Accuracy (0.5227) is slightly higher than Model A's in cross-testing.
- The model seems to adapt slightly better to balanced datasets despite being trained on imbalanced data.

- High recall for majority classes from training data carries over to balanced test data (Class 5: Recall 0.79).
- Lower recall for underrepresented classes in original training (Class 2: Recall 0.35).

Confusion Matrix:

```
[[6605 2203  811  179  202]
 [3515 3537 2077  599  272]
 [1127 2026 3669 2442  736]
 [ 216  488 1625 4460 3211]
 [ 168  134  359 1475 7864]]
```

Observations

- Model A is more balanced in performance across classes but struggles when class distribution shifts heavily.
- Model B maintains higher performance for originally frequent classes, even when tested on balanced data.
- Both models show limitations in adaptability, but Model B appears slightly more robust in cross-distribution scenarios.

Based on the results:

Cross-testing confirms that data distribution in training strongly affects performance under distribution shifts.

Balanced training improves fairness but may suffer in highly skewed scenarios.

Imbalanced training favors majority classes but risks low performance for minority classes when tested in balanced environments.

For deployment, a hybrid approach such as balanced training with weighted loss could help improve robustness.